AMR EDGE COURSE – MODULE D

Bioinformatics Analysis for Genomic AMR Surveillance

Program Outline

DAY 1	ACTIVITY
08:30 - 09:00	**Registration and Introductions** Opening remarks by Prof Beverly Egyir
09:00 – 10:00	**Theory** Introduction to AMR **Prof Beverly Egyir**
10:00- 10: 20	**Theory** Introduction to DNA extraction and Workflow on sequencing platforms **Grebstad Amuasi**
10:20- 10:30	Break
10:30 – 10:50	**Theory** Introduction to Bioinformatics and platforms for analyses **Christian Owusu-Nyantakyi**
10:50 – 11:10	**Theory** Data management & WHONET **Alfred Bortey**
11:10- 11:30	**Theory** **Guest Lecture**
11:30- 12:00	**Group photos| Grouping & Interactions with facilitators**
12:00 - 13:00	Lunch
13:00 - 16:30	**Theory/Practicals** Introduction to Linux *Terminal Navigation | File formats | Basic commands | Bash scripting* **Christian, William, Grebstad**
16:30 - 17:00	**Wrap-up & Discussion**

DAY 2	ACTIVITY
08:30 – 10:30	**Practical** Quality Control of Sequenced Reads *Quality assessment- FastQC, MultiQC, NanoPlot, NanoQC|Trimming - Trimmomatic, Fastp* **William | Christian, Grebstad**
10:30 - 10:45	Break
10:45 - 12:00	**Theory** Introduction to Genome Assembly **Christian** **Practical** De novo Assembly – Illumina, ONT

	Unicycler | Flye **Christian | William, Grebstad**
12:00 - 13:00	Lunch
13:00 – 14:30 14:30 – 16:30	**Practical** Reference Based Assembly *BWA | Minimap | Samtools | bcftools* **William |Grebstad, Christian** **Practical** Demonstration of utility of the rMAP pipeline **Dr. Gerald Mbowa**
16:30 - 17:00	**Practical** Assembly Quality Assessment - Completeness, Contamination *Quast | Checkm* **William |Grebstad, Christian**
DAY 3	**ACTIVITY**
08:30 - 08:50	**Practical** Genome Submission *NCBI* **William |Grebstad, Christian**
08:50 – 09:15	**Theory** Introduction to Resistance and Virulence Determinants and Plasmid Detection **Grebstad**
09:00 – 10:30	**Practical** AMR Gene Detection Tools *ResFinder | CARD| Pathogen watch* **Grebstad | William, Christian**
10:30 - 10:45	Break
10:45 – 11:30	**Practical** AMR Gene Detection Tools *AMRFinder Plus | Abricate* **Christian | William, Grebstad**
11:30 - 13:00	**Practical** Pathogenicity and Virulence Factor Prediction *Virulence Finder | Pathogen watch |Pathogen Finder 2* **William | Grebstad, Christian**
13:00 - 14:00	Lunch
14:00 - 16:30	**Practical** MGE Prediction Tools *Plasmid Finder | MOB-suite | MGE-Finder* **William | Grebstad, Christian**

16:30 - 17:00	Wrap up & Discussion
DAY 4	**ACTIVITY**
08:30 – 09:30	**Theory** Molecular Typing of Bacterial Species William
09:30 - 10:30	**Practical** Molecular Typing of Bacterial Species *MLST| cgMLST|Seqsero |Kleborate|Serotypefinder | PAst* **Christian | William, Grebstad**
10:30 - 10:45	Break
10:45 - 11:30	**Theory** Phylogenetics **Grebstad**
11:30 - 13:00	**Practical** Phylogenetics and Tree Visualization *CSI-Phylogeny | Bactinspector |Snippy | IQ-Tree | IToL* **Christian| William | Grebstad**
13:00 - 14:00	Lunch
14:00 - 15:00	**Practical** Outbreak Investigation **Grebstad | Christian | William**
15:00 - 17:00	Group Discussions
DAY 5	**ACTIVITY**
08:30 - 09:00	Recap of Day 4
9:00- 10:40	Group Presentations
10:40 – 12:00	**Closing Ceremony** *Presentation of Certificates| Closing Remarks| Group Photos| Departure*